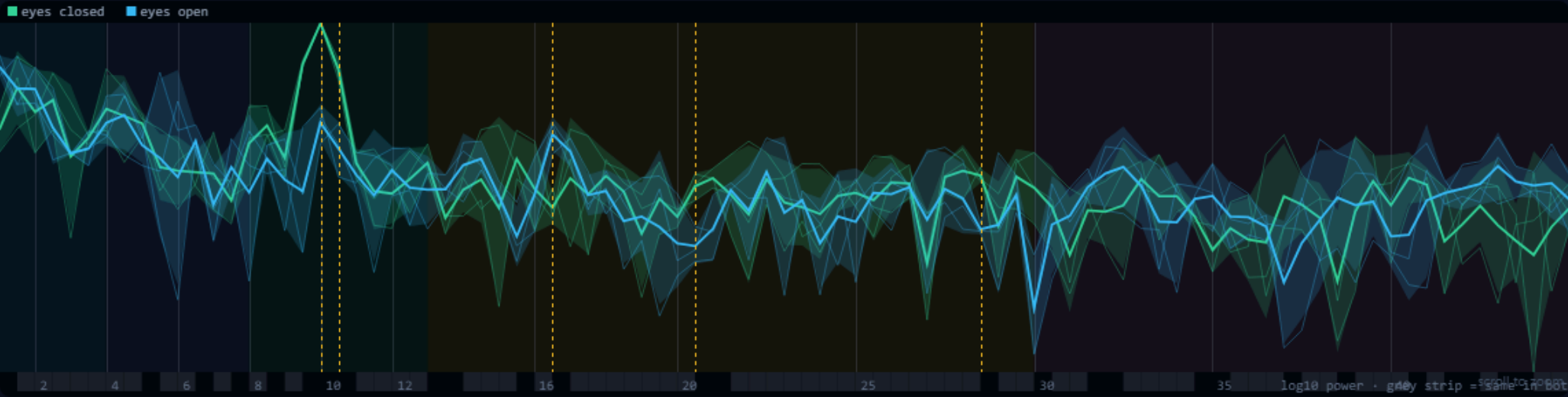


~\$1,068/year
to read raw data from
a headset you own.

So the dongle encrypts its own output to enforce it.

I reverse-engineered it instead.

SPECTRA, WITH EACH STATE'S SPREAD A mean alone invites reading a difference into curves that are inside each other's band. The band is the honest part.



DIFFERENCE PER FREQUENCY BIN eyes closed vs eyes open measured in units of each state's own spread, so 1.0 means "as big as the scatter"



IS THE BIGGEST DIFFERENCE MORE THAN LUCK?

Largest difference **8.45** at **28.5 Hz**.

Shuffling the state labels reaches that or more in **8.5%** of 200 trials (chance alone reaches **8.45**).

COULD A SIMPLE RULE TELL THEM APART?

Nearest-centroid, leaving each state out in turn, using the 5 most different frequency bins: **100%** of 6 states correct.

Using all 89 bins instead: **67%**. Chance is 50%. 6 states is a handful — this is a hint about which frequencies carry the difference, not a validated classifier.
Bins used: 28.5, 10, 16.5, 10.5, 20.5 Hz

WHAT BOTH STATES HAVE IN COMMON

62 of 89 frequency bins (**70%**) are the same in both: the difference there is smaller than the states' own spread.

Those bins cannot explain the difference, however much the two curves appear to wiggle in them. If a promising-looking band is listed here, it is not the signal. Wholly shared bands: theta, gamma.

- 17.5 Hz
- 1.5 Hz
- 4.5 Hz
- 29 Hz
- 38 Hz
- 22 Hz
- 3 Hz
- 6 Hz
- 11.5 Hz
- 39 Hz

where they differ	Hz	effect	eyes closed	eyes open
differs	28.5	8.45	4.62	3.95
differs	10	7.56	6.51	5.28
differs	16.5	4.36	4.22	5.13
differs	10.5	3.90	5.94	4.96
differs	20.5	3.83	4.49	3.74
differs	1	3.57	5.10	5.08

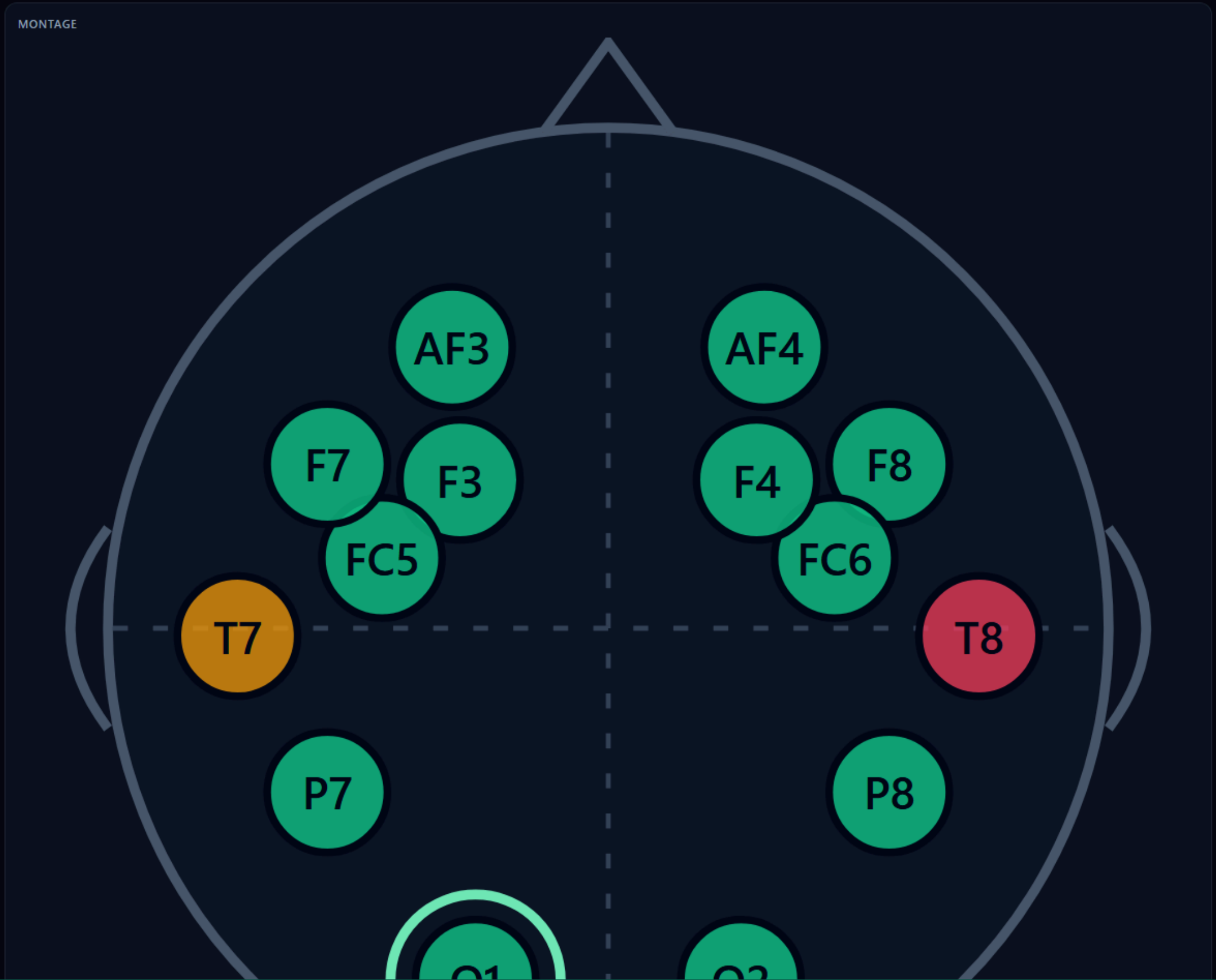
electrode	largest effect	at
P8	12.95	10.5 Hz
T7	10.15	10 Hz
O1	8.45	28.5 Hz
AF4	6.19	20 Hz
F8	6.15	27.5 Hz
O2	5.52	9.5 Hz

It measures. It doesn't cheerlead.

Effect size per frequency bin, the 70% of bins both states share, and a p-value that states the floor it cannot beat.

synthetic source · the app labels it demo

SIGNAL		REPORTS/S	SAMPLES	ELAPSED	KEY	CONTACT
Demo signal	synthetic signal (no headset)	151	542815	73:37	n/a	12/14 good



14 channels of raw EEG, live.

No Emotiv software, no account, no licence — straight off the dongle at 128 Hz, with a contact map and a live O1/O2 alpha meter.

synthetic source · the app labels it demo

Eyes closed / eyes open

SHAPE

Linear — each state once, in order

Loop — repeat until you stop

START WITH (COUNTDOWN BEFORE RECORDING)

3

seconds, spoken
3-2-1

REPEAT (BLANK = UNTIL STOPPED)

until stopped

REST BETWEEN REPETITIONS (SECONDS, UNLABELLED)

0

DISCARD FROM THE END WHEN STOPPED

2

NOTES

Alternates eyes-closed and eyes-open. Alpha (8-12 Hz) rises over the occipital channels with the eyes closed; that contrast is the signal.

States

In order. Leave the duration blank for "until I stop" — allowed only as the last state.

+ Add state

	NAME (ALSO SPOKEN)	SECONDS	SAY INSTEAD	
1	eyes closed	20	close your eyes	☐ until stopped ↑ ↓ ×
2	eyes open	15	open your eyes	☐ until stopped ↑ ↓ ×

What it will actually do

open-ended

a manual stop drops the last 2 state(s)

This protocol is valid.

get ready 3s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

eyes open 15s

eyes closed 20s

... and on

Raw EEG read directly from the dongle — no Emotiv software or licence. Recordings are biometric data: they stay in recordings/, which is git-ignored.

Tell it what to do. It runs the experiment.

eyes closed 20 s → eyes open 15 s, repeat — labelled as it goes, with the run order previewed before you save.

synthetic source · the app labels it demo



STARTING WITH

3

Nothing is being recorded yet. The countdown is narrated and left out of the dataset — recording begins the moment it reaches zero.

LIVE CHECK



01
alpha 49.9% · peak 6.93

02
alpha 57.9% · peak 6.72

With the eyes closed, alpha should rise here. That is the check that the recording is worth keeping.

3, 2, 1 — spoken out loud.

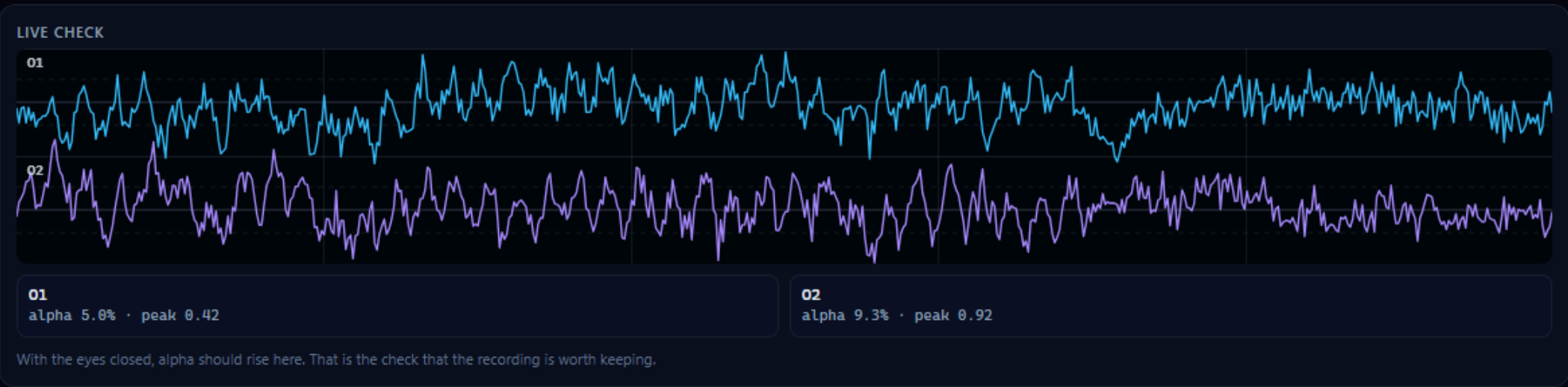
The countdown isn't data, so the file is created the moment it reaches zero. Stop during it and you get no dataset at all — because there was nothing to record.

synthetic source · the app labels it demo

subject, electrode condition, anything unusual

☒ Speak the states and the countdown

▶ Start protocol



Dataset written

eyes closed

DURATION

7 s

SAMPLES

909

KEPT

eyes closed

[Open in Datasets →](#)

1x · 7 s

Raw EEG read directly from the dongle — no Emotiv software or licence. Recordings are biometric data: they stay in recordings/, which is git-ignored.

Dataset written. Labelled. Sample-exact.

Boundaries are queued with a stream time and applied by the writer as it passes them.

synthetic source · the app labels it demo

LIBRARY

6

Refresh

eyes closed

7 s · 909 samples · 100 kB

● eyes closed 7 s

2026-09-15T14:00:16

labelled dataset



eyes closed

4 s · 536 samples · 60 kB

● eyes closed 4 s

2026-09-14T22:00:42

labelled dataset



eyes closed

2 s · 211 samples · 25 kB

● eyes closed 2 s

2026-09-14T21:59:57

labelled dataset



showcase eyes closed vs open

17 s · 2221 samples · 246 kB

● eyes open 9 s ● eyes closed 9 s

2026-09-14T21:59:30

labelled dataset



Eyes closed / eyes open

4 m 05 s · 31338 samples · 3.4 MB

● eyes closed 2 m 20 s ● eyes open 1 m 45 s

2026-09-13T14:56:07

labelled dataset



live_2026-09-13_14-51-40

15 s · 1896 samples · 201 kB

2026-09-13T14:51:40

live_view session



Overview

Discovery

showcase eyes closed vs open

labelled dataset

from demo

▶ Replay

 Delete

DURATION

17 s

SAMPLES

2221

STATES

6

SIZE

246 kB

C:\Users\titus\projects\brain-waves-projects\recordings\showcase-eyes-closed-vs-open_2026-09-14_21-59-30

STATES, IN PROPORTION

Click a state to show only its window below — and to take it to Discovery.



● eyes open · 3× · 9 s ● eyes closed · 3× · 9 s

Every state, in proportion.

Click a state and every chart narrows to its window. Four files: eeg.csv, labels.csv, meta.json, events.jsonl.

synthetic source · the app labels it demo

Neither bug was the crypto.

1 · It picked the wrong key

emokit chooses its AES key with

`serial.startswith("UD2016")` — a literal string test. My dongle is a 2018 unit, so it silently took the wrong key and printed confident garbage. No exception. No warning.

2 · It read the bytes backwards

Every 16-bit sample field was decoded **big**-endian. They are **little**-endian. Same offsets, opposite order — which is why the output still looked like a signal for eight years.

A wrong key never throws.

It produces beautiful, plausible, meaningless numbers. So nothing here is claimed without a check a wrong key cannot pass:

1 Byte 1 must collapse

Correct: 2 distinct values (0x10 / 0x20). Wrong: ~256.

2 Autocorrelation

Real EEG is band-limited and correlated in time. Noise is not.

little-endian, EEG packets

+0.797

big-endian — what emokit did

+0.480

wrong key

+0.007

AND THEN THE TEST THAT MATTERS · **CONFIRMED**

It is a real brain.

The eyes-closed alpha rhythm — an 8–12 Hz peak over the occipital channels that rises when the eyes close and drops when they open.

the peak rose with the eyes closed

on O1 and on O2

the measured values

every pair

consistent

withheld

Seven pairs, wet pads, real dongle — not the demo. The numbers are derived from my own brain activity, so they stay private.

A wrong key produces beautiful charts too — which is exactly why this check existed.

What would you build on it?



Code, MIT, credited to emokit

github.com/titusfx/brain-waves-projects



Why you're looking at a synthetic head

Every screenshot in this deck is the app's labelled demo source. My real session is biometric data, so it stays private — the numbers are what I'm willing to publish.